Wentao Zhang

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Education

Southeast University 2021/09 - 2025/09
Ph.D. in Electrical Engineering Nanjing, China

- Dissertation, "Multi-physics Analysis and Design of Hairpin Winding PM Traction Machines in Electric Vehicles". Doctoral supervisor Prof. Zhongze Wu.
- **Industrial prototype development with Inovance** (world-leading EV traction machine supplier): technical lead for two high-speed oil-cooled PM traction machines taken through the full design-build-test cycle. **20,000 rpm, A-class:** 270 kW, 380 Nm, 7.87 kW/kg, 97.4% peak efficiency. **18,000 rpm, A0-class:** 113 kW, 138 Nm, 7.78 kW/kg, 95.3% peak efficiency (all measured).
- **Design:** multi-objective optimisation coupling fast index models, electromagnetic FEA, thermal networks and stress evaluation. **Manufacture:** engineering drawing and tolerance release; coordination of external suppliers for machining, lamination stacking, hairpin forming and assembly. **Test:** AVL dynamometer characterisation (efficiency maps, thermocouple and waveform data) used to validate the design models.
- 2025 MDPI *Machines* Best PhD Thesis Award (awarded for this doctoral thesis).

Southeast University 2018/09 - 2021/06
M.Eng. in Electrical Engineering Nanjing, China

- Dissertation, "Analysis and Design of High-Performance Wound-Excited Magnetic Field Modulation-based Brushless Machines".
- 1st Prize, IEEE IAS CMD Thesis Contest, 2023 (awarded for this master's thesis).
- Outstanding Master Degree Thesis Award, Jiangsu Province, 2022.

Southeast University 2014/09 - 2018/06
B.Eng. in Electrical Engineering Nanjing, China

- Dissertation, "Comparative Study of Control Systems for PM Machine in Electric Vehicles".

Employment

University of Nottingham 2025/09 - present
Research Assistant, Power Electronics and Machines Centre (PEMC) Nottingham, UK

- Within the EPSRC-funded AIMD programme (Nottingham, Bristol, Newcastle), I work on next-generation magnet-free synchronous machines, develop an in-house finite-element code package, and extend operator-learning neural surrogates to accelerate machine design.

Research Expertise

- **Areas.** Multiphysics modelling and simulation of electrical machines (electromagnetic, thermal, fluid, and structural-mechanical); digital twins and reduced-order modelling; physics-informed and operator-learning AI / machine learning; high-speed electromechanical machine design; multi-objective design optimisation; electric drives and power electronics; design for manufacture.
- **Computational methods developed.** In-house finite-element code (electromagnetic, structural-modal, mechanical-stress); deep-learning and surrogate modelling in PyTorch with GPU computing; lumped-parameter thermal-network modelling.
- **Engineering software.** ANSYS Maxwell 2D/3D, JMAG, FEMM 4.2, Motor-CAD, ANSYS Mechanical, MATLAB / Simulink, SolidWorks, AutoCAD.
- **Programming.** Python (primary), C / C++ (TI C2000, STM32 real-time control), MATLAB, Rust.
- **Experimental and hardware.** AVL dynamometer testing and efficiency-map extraction, multi-thermocouple instrumentation, three-phase converter prototyping, manufacturing-drawing and tolerance release.

Awards and Honours

- 2025 **1st Prize, IEEE TEC Electric Vehicle Design Competition**.
- 2025 **2nd Prize Paper Award, IEEE Transactions on Transportation Electrification** (for first-author paper [J5]).

- 2025 **MDPI *Machines* Best PhD Thesis Award**.
- 2024 2nd Prize, 11th Energy Equipment Innovative Design Competition for China Postgraduate.
- 2023 CIEEC Best Paper Award.
- 2023 1st Prize, IEEE IAS CMD Thesis Contest.
- 2023 Wiscom System Scholarship.
- 2023 1st Prize Academic Scholarship.
- 2022 Outstanding Master Degree Thesis Award, Jiangsu Province.

Publications

My record comprises **26 journal papers** (20 in IEEE Transactions; 11 first-author), **19 peer-reviewed conference papers** (10 first-author), and **9 patents**, on multiphysics modelling, digital twins, and AI-accelerated design of electro-mechanical machines.

Journal Articles (26 papers, 20 in IEEE Transactions)

[J1] **Wentao Zhang**, W. Hua, Z. Wu, G. Zhao, Y. Wang, and W. Xia, "Analysis of DC Winding Induced Voltage in Wound-Field Flux-Switching Machine With Air-Gap Field Modulation Principle," *IEEE Trans. Ind. Electron.*, vol. 69, no. 3, pp. 2300–2311, Mar. 2022.

[J2] **Wentao Zhang**, Z. Wu, W. Hua, Z. Zhu, Z. Wu, J. Hu, et al., "Reduction of Open-Circuit DC Winding Induced Voltage and Torque Pulsation in the Wound Field Switched Flux Machine by Stator Axial Pairing of Tooth Tips," *IEEE Trans. Ind. Appl.*, vol. 58, no. 2, pp. 1976–1990, Mar.–Apr. 2022.

[J3] **Wentao Zhang**, Z. Wu, Y. Fan, W. Hua, and M. Cheng, "A Stable and Computationally Efficient Spatial Harmonic Model for Predicting the DC Winding Induced Voltage in WFSF Machine," *IEEE Trans. Ind. Appl.*, vol. 59, no. 4, pp. 3966–3977, Jul.–Aug. 2023.

[J4] **Wentao Zhang**, Z. Wu, Y. Fan, W. Hua, and M. Cheng, "Analysis and Improvement of DC Winding Current Dynamics in Wound Field Switched Flux Machine," *IEEE Trans. Ind. Appl.*, vol. 60, no. 4, pp. 6023–6032, Jul.–Aug. 2024.

[J5] **Wentao Zhang**, Z. Wu and W. Hua, "Design of *n*-Layer Flat Wire Winding in Permanent Magnet Machine for Electric Vehicles," *IEEE Trans. Transport. Electrific.*, vol. 11, no. 3, pp. 7798–7810, 2025. (2025 IEEE TTE Prize Paper Award, 2nd Prize.)

[J6] **Wentao Zhang**, Z. Wu, L. Jin, Y. Fan, W. Hua, and M. Cheng, "Analysis and Multi-Objective Optimization of the Hybrid Excitation Switched Flux Machine," *IEEE Trans. Ind. Appl.*, vol. 61, no. 2, pp. 2996–3006, Mar.–Apr. 2025.

[J7] **Wentao Zhang**, Z. Wu, Z. Wang, W. Hua, Y. Fan, Z. Xu, et al., "Design Tradeoff Between Flux Regulation Capability and DC Winding Induced Voltage in Hybrid Excitation Switched Flux Machine," *IEEE Trans. Ind. Appl.*, vol. 61, no. 5, pp. 7198–7208, Sep. 2025.

[J8] **Wentao Zhang**, Y. Fan, Z. Zhu, Z. Wu, W. Hua, and M. Cheng, "Analysis of DC Winding Induced Voltage in Wound-Rotor Synchronous Machines by Using the Air-Gap Field Modulation Principle," *World Electr. Veh. J.*, vol. 13, no. 11, Art. no. 215, 2022.

[J9] **Wentao Zhang**, Z. Wu, Z. Cang, W. Hua, Y. Fan, and M. Cheng, "Influence of End-Effect in Switched Flux Machines," *IEEE Trans. Ind. Appl.*, vol. 62, no. 1, pp. 785–798, Jan. 2026.

[J10] **Wentao Zhang**, Z. Wu and W. Hua, "High-Fidelity Lumped Parameter Thermal Model for Oil-Cooling Hairpin Winding PM Machines in EVs," *IEEE Trans. Transport. Electrific.*, vol. 12, no. 2, pp. 2978–2991, Apr. 2026.

[J11] **Wentao Zhang**, Z. Wu and W. Hua, "Design of Notching and Mechanical Stress Relieving Grooves for High-Speed PM Machine in EVs," *IEEE Trans. Ind. Appl.*, early access, 2026.

[J12] Z. Wu, **Wentao Zhang**, Y. Fan, W. Hua, and M. Cheng, "A Transient-State Lumped Parameter Thermal Model for Brushless Wound Field Switched Flux Machines," *IEEE Trans. Transport. Electrific.*, vol. 10, no. 1, pp. 224–238, Mar. 2024.

[J13] Z. Wu, Z. Zhu, C. Wang, W. Hua, K. Wang, and **Wentao Zhang**, "Influence of Rotor Iron Bridge Position on DC-Winding-Induced Voltage in Wound Field Switched Flux Machine Having Partitioned Stators," *Chin. J. Electr. Eng.*, vol. 7, no. 3, pp. 20–28, Sep. 2021.

[J14] Z. Wu, Z. Zhu, S. Cai, W. Hua, and **Wentao Zhang**, "Enhancement of Torque Density in Wound Field Switched Flux Machines With Partitioned Stators Using Assisted Ferrites," *Chin. J. Electr. Eng.*, vol. 7, no. 3, pp. 42–51, Sep. 2021.

[J15] Z. Wu, L. Jin, W. Hua, **Wentao Zhang**, and M. Cheng, "Harmonics Orders Modeling of DC Winding Induced Voltage Pulsation in Wound Field Switched Flux Machines Under PWM Excitation," *IEEE Trans. Transport. Electrific.*, vol. 10, no. 3, pp. 7040–7050, Sep. 2024.

[J16] Z. Wu, L. Jin, **Wentao Zhang**, Y. Fan, W. Hua, and M. Cheng, "Influence of PWM Excitation on DC Winding Induced Voltage Pulsation in Wound Field Switched Flux Machines," *IEEE Trans. Ind. Appl.*, vol. 60, no. 1, pp. 460–476, Jan.–Feb. 2024.

[J17] Z. Wu, Z. Zhu, W. Hua, S. Akehurst, X. Zhu, **Wentao Zhang**, et al., "Analysis and Suppression of Induced Voltage Pulsation in DC Winding of Five-Phase Wound-Field Switched Flux Machines," *IEEE Trans. Energy Convers.*, vol. 34, no. 4, pp. 1890–1905, Dec. 2019.

[J18] L. Jin, W. Hua, U. B. Akuru, Z. Wu, **Wentao Zhang**, and M. Cheng, "Comparative Study on DC Winding Induced Voltage Pulsation of Wound Field Flux Modulation Machines Having Different Iron Core Structures," *IEEE Trans. Ind. Appl.*, vol. 61, no. 2, pp. 2874–2883, Mar.–Apr. 2025.

[J19] X. Yan, Z. Wu, W. Hua, **Wentao Zhang**, and M. Cheng, "A Mathematical Model for Wound Field Switched Flux Machine Considering Inductance Harmonics of Field and Armature Windings," *IEEE Trans. Ind. Appl.*, vol. 60, no. 5, pp. 6811–6822, Sep.–Oct. 2024.

[J20] H. Hua, X. Chen, D. Li, **Wentao Zhang**, Z. Wu, and W. Hua, "Investigation of Torque Improvement of Modular Stator Surface-Mounted Permanent Magnet Machines With Flux Gaps," *IEEE Trans. Ind. Appl.*, vol. 60, no. 5, pp. 6787–6798, Sep.–Oct. 2024.

[J21] J. Chen, W. Hua, L. Shao, Z. Wu, X. Yan, and **Wentao Zhang**, "Modified Magnetic Equivalent Circuit of Double-Stator Single-Rotor Axial Flux Permanent Magnet Machine Considering Stator Radial-End Flux-Leakage," *IET Electr. Power Appl.*, vol. 18, no. 2, pp. 195–207, Feb. 2024.

[J22] Z. Wu, X. Yan, Z. Wang, **Wentao Zhang**, and W. Hua, "Influence of PWM Excitation on Loss and Temperature of Flat Wire Permanent Magnet Machines for Electric Vehicles," *IEEE Trans. Ind. Electron.*, vol. 73, no. 4, pp. 5174–5185, Apr. 2026.

[J23] Z. Wu, E. Fu, J. Du, **Wentao Zhang**, and W. Hua, "Quantitative Comparison of Electromagnetic Vibration of PMSMs With Glued and Welded Stator Core for Electric Vehicles," *IEEE Trans. Ind. Appl.*, early access, 2025.

[J24] Z. Wu, **Wentao Zhang**, and W. Hua, "Design, Analysis and Performance Evaluation of Hairpin Winding Permanent Magnet Machines," *IEEE Trans. Transport. Electrific.*, accepted for publication, 2026.

[J25] Z. Cang, **Wentao Zhang**, Z. Wu, and W. Hua, "Open-Circuit Magnetic Equivalent Circuit Model for Single-Side Double-Row Stator Axial Flux Wound Field Machine," *IET Electr. Power Appl.*, vol. 20, no. 1, Art. no. e70216, 2026.

[J26] Z. Wang, L. Jin, Z. Wu, W. Hua, **Wentao Zhang**, and M. Cheng, "Suppression of DC Winding Current Harmonics in Wound-Field Machines Under PWM Excitation by Using Active Voltage Injection and LC Low-Pass Filters," *IET Electr. Power Appl.*, vol. 20, no. 1, Art. no. e70157, 2026.

Conference Papers (19 papers)

[C1] **Wentao Zhang**, Z. Wu and W. Hua, "A Deep Learning Based Magnetic Field Surrogate Modeling Method for Interior Permanent Magnet Machine in EVs," in *Proc. Int. Conf. Elect. Mach. (ICEM)*, Funchal, Madeira, Portugal, 2026. (Accepted.)

[C2] **Wentao Zhang**, Z. Wu and W. Hua, "Comparative Investigation of Bridge-Retained and Carbon Fiber Sleeved Interior Permanent Magnet Machines in Electric Vehicles," in *Proc. Int. Conf. Elect. Mach. Syst. (ICEMS)*, Singapore, 2026. (Accepted.)

[C3] **Wentao Zhang**, Z. Wu, W. Hua and Z. Zhu, "Reduction of Open-Circuit DC Winding Induced Voltage and Torque Pulsation in the Wound Field Switched Flux Machine by Stator Axial Pairing of Tooth-Tips," in *Proc. IEEE Int. Elect. Mach. Drives Conf.*, Gothenburg, Sweden, 2020, pp. 522–528.

[C4] **Wentao Zhang**, Z. Wu, Y. Fan, W. Hua and M. Cheng, "A Stable and Computationally Efficient Spatial Harmonic Model for Predicting the DC Winding Induced Voltage in WFSF Machine," in *Proc. IEEE Energy Convers. Congr. Expo.*, Detroit, MI, USA, 2022, pp. 1–7.

[C5] **Wentao Zhang**, Z. Wu, W. Hua, Y. Fan, Z. Xu and M. Cheng, "Design Tradeoff between Flux Regulation Capability and DC Winding Induced Voltage in Hybrid Excitation Switched Flux Machine," in *Proc. IEEE Energy Convers. Congr. Expo.*, Nashville, TN, USA, 2023, pp. 5060–5066.

[C6] **Wentao Zhang**, Z. Wu, L. Jin, Y. Fan, W. Hua and M. Cheng, "Analysis and Multi-Objective Optimization of the Hybrid Excitation Switched Flux Machine," in *Proc. Int. Elect. Energy Conf.*, Hefei, China, 2023, pp. 3371–3376. (Best Paper Award.)

[C7] **Wentao Zhang**, Z. Wu, W. Hua, Y. Fan and M. Cheng, "Influence of the End-Effect in Wound Field Switched Flux Machine," in *Proc. Int. Conf. Elect. Mach. Syst.*, Zhuhai, China, 2023, pp. 1223–1227.

[C8] **Wentao Zhang**, Z. Wu, Y. Fan, W. Hua and M. Cheng, "Analysis and Improvement of DC Winding Current Dynamics in Wound Field Switched Flux Machine," in *Proc. Int. Conf. Energy Technol. Future Grids*, Wollongong, Australia, 2023, pp. 1–6.

[C9] **Wentao Zhang**, Z. Wu, W. Hua and J. Qi, "Analysis of PM Volume Reduction Methods for Hybrid Excitation Switched Flux Machines," in *Proc. Int. Conf. Elect. Mach. Syst.*, Zhuhai, China, 2024, pp. 3152–3157.

[C10] **Wentao Zhang**, Z. Wu and W. Hua, "Field-Oriented Design of Notching and Mechanical Stress Relieving Grooves for Rotors in Flat Wire Permanent Magnet Machines," in *Proc. IEEE Int. Elect. Mach. Drives Conf.*, 2025, pp. 304–309.

[C11] Z. Wu, L. Jin, **Wentao Zhang**, Y. Fan, W. Hua and M. Cheng, "Influence of PWM Excitation on DC Winding Induced Voltage in Wound Field Switched Flux Machines," in *Proc. IEEE Energy Convers. Congr. Expo.*, Detroit, MI, USA, 2022, pp. 1–7.

[C12] L. Jin, W. Hua, U. B. Akuru, Z. Wu, **Wentao Zhang** and M. Cheng, "Comparative Study on DC Winding Induced Voltage Pulsation of Wound Field Flux Modulation Machines Having Different Iron Core Structures," in *Proc. Int. Conf. Elect. Mach. Syst.*, Zhuhai, China, 2023, pp. 442–447.

[C13] M. P. Koroma, W. Hua, Z. Wu, L. Jin and **Wentao Zhang**, "Influence of Magnet Layer Numbers on Electromagnetic Performance of Interior Permanent Magnet Machines," in *Proc. Int. Conf. Elect. Mach. Syst.*, Zhuhai, China, 2023, pp. 696–700.

[C14] X. Chen, H. Hua, D. Li, **Wentao Zhang**, Z. Wu and W. Hua, "Modular Stator Surface-Mounted Permanent Magnet Machines with Flux Gaps," in *Proc. IEEE Int. Elect. Mach. Drives Conf.*, San Francisco, CA, USA, 2023, pp. 1–7.

[C15] X. Yan, Z. Wu, W. Hua, **Wentao Zhang** and M. Cheng, "A Mathematical Model for Wound Field Switched Flux Machine Considering Inductance Harmonics of Field and Armature Windings," in *Proc. IEEE Int. Elect. Mach. Drives Conf.*, San Francisco, CA, USA, 2023, pp. 1–6.

[C16] Z. Wu, L. Jin, **Wentao Zhang**, X. Yan, H. Ge, H. Hua, W. Hua and M. Cheng, "A Lumped Parameter Thermal Model for Variable Flux Reluctance Machine Considering Relative Position of DC and AC Windings," in *Proc. Int. Conf. Elect. Mach.*, Torino, Italy, 2024, pp. 1–7.

[C17] Z. Wu, X. Yan, Z. Wang, **Wentao Zhang** and W. Hua, "Quantitative Analysis of the Torque-Speed Curve for a Flat Wire Permanent Magnet Machine Drive System," in *Proc. IEEE Int. Elect. Mach. Drives Conf.*, 2025, pp. 40–44.

[C18] Z. Wu, E. Fu, J. Du, **Wentao Zhang** and W. Hua, "Influence of Stator Core on the Vibration Behavior of Permanent Magnet Machines with Hairpin Winding," in *Proc. IEEE Int. Elect. Mach. Drives Conf.*, 2025, pp. 34–39.

[C19] Z. Wu, **Wentao Zhang**, Z. Wang, L. Shao, W. Hua and M. Cheng, "Stator Semi-Skewing Method for High-Speed Permanent Magnet Machine," in *Proc. Int. Conf. Elect. Mach. Syst.*, 2025, pp. 693–697.

Patents (9)

[P1] Z. Wu, **Wentao Zhang**, L. Jin, and W. Hua, "Inductive Ripple Voltage Suppression Method Based on Low-Pass Filter Circuit," CN118508832A, 2024.

[P2] Z. Wu, **Wentao Zhang**, L. Jin, and W. Hua, "Disk Brushless Electric Excitation Magnetic Field Modulation Motor With Double-Row Stator Structure," CN118100572A, 2024.

[P3] Z. Wu, **Wentao Zhang**, H. Huang, and W. Hua, "Method for Constructing a High-Precision Motor Performance Database Based on a Dual-Model Strategy," CN117972924A, 2024.

[P4] Z. Wu, **Wentao Zhang**, W. Hua, and Z. Zou, "Method for Establishing General Lumped Parameter Thermal Network Model of Motor," CN117094088A, 2023.

[P5] Z. Wu, **Wentao Zhang**, L. Jin, and W. Hua, "Method for Analyzing Harmonic Orders of Induced Ripple Voltage of Excitation Winding," CN115642847A, 2023.

[P6] Z. Wu, **Wentao Zhang**, J. Liao, Y. Fan, W. Hua, and M. Cheng, "Excitation Winding Induced Pulse Voltage Suppression Method Based on Harmonic Current Injection," CN115208245A, 2022.

[P7] Z. Wu, **Wentao Zhang**, Y. Fan, W. Hua, and M. Cheng, "High-Precision Field Path Coupling Design Method for Brushless Electric Excitation Magnetic Field Modulation Motor," CN114781227A, 2022.

[P8] Z. Wu, **Wentao Zhang**, W. Hua, and Z. Zhu, "Method for Weakening the Induced Pulsating Voltage of a Brushless Electric Excitation Magnetic Field Modulated Motor," CN114189185A, 2022.

[P9] W. Hua, **Wentao Zhang**, Z. Wu, and S. Ding, "Method for Reducing Torque Ripple and Induced Voltage of Electric Excitation Flux Switching Motor," CN111769697A, 2020.

Research Funding

- **N-ZEEE Flexible Funding award** (Net-Zero Enabling Electrical Engineering network, 2026), **£2,700**, approved by the network's selection panel on my proposal, to present [C2] at IEEE ICEMS 2026 (Singapore, 24 to 27 November 2026).

- **LLM-agent-assisted design and training of physics-informed surrogate models** (PINNs and Transformers) for multiphysics PDE simulation: an ongoing six-month project I lead (2026), supported by in-kind API credits from the **OpenAI Researcher Access Program**.
- Contributed as a **participating researcher** to externally funded programmes, including the EPSRC **AIMD** programme at Nottingham (2025 to present) and the **Inovance**-funded industrial traction-machine development programme at Southeast University (2021 to 2025).

Research Supervision and Mentoring

- Provided day-to-day technical guidance to junior PhD and M.Eng. researchers in the SEU electrical-machines group, in multiphysics modelling, finite-element analysis, thermal-network construction and PWM-loss coupling, leading the technical formulation on multiple jointly authored papers, as reflected in the co-authorship of the publication list.
- **Technical lead** on the two-prototype Inovance development programme, coordinating multiphysics design, manufacturing-drawing release, and dynamometer testing.
- Continuing informal mentoring of junior research students at PEMC, University of Nottingham.

Teaching

- **Teaching Assistant**, postgraduate course "Integrated Practice of Electrical Machine Systems," Southeast University (2021 to 2025), covering tutorial preparation and delivery, Q&A and consultation hours, structured written feedback on student final-project proposals, and mark recommendations to the course coordinator.
- Developed tutorial and project-assessment materials for the above course, and research-derived teaching resources (a coupled electromagnetic-thermal modelling exercise and an embedded motor-drive practical).

Professional Service

- **Referee** for IEEE Transactions journals, including *IEEE Trans. Industry Applications*, *IEEE Trans. Energy Conversion*, and *IEEE Trans. Transportation Electrification*.
- **Co-Organiser**, ICEM 2026 Special Session "Next-Generation Electrified Transport: Advances in Electric Drive Systems," Funchal, Portugal, Sep. 2026 (with Dr Lingyun Shao, NUAA).

Research Impact and Translation

- **Internationally recognised research.** My first-author paper on *n*-layer flat-wire winding design [J5] received the **2025 IEEE Transactions on Transportation Electrification Prize Paper Award (2nd Prize)**, a competitively selected recognition among the best papers in one of the field's leading journals. In the same year my machine-design entry won **1st Prize in the IEEE TEC Electric Vehicle Design Competition**, an international design contest organised by the IEEE Transactions on Energy Conversion.
- **Innovation and technology translation.** 9 patents, and two high-speed oil-cooled traction-machine prototypes developed with **Inovance** and taken through the full design-build-test cycle (measured up to 270 kW, 7.87 kW/kg, and 97.4% peak efficiency at 20,000 rpm), contributing to energy-efficient transport electrification.

Additional Information

- **Languages.** English (full professional proficiency; the medium of all 45 peer-reviewed publications and of my current employment in the UK), Mandarin Chinese (native).
- Cross-cultural research and engineering experience across China (Southeast University) and the United Kingdom (University of Nottingham), supporting international and industrial collaboration.

Last Updated on October 20, 2026